# Do you know V=IR ?

A VR Engineering Mystery with Real-World Object Recognition

Course: Mechatronics and XR Lab (2026) | Project Type: VR × Mechatronics Integration
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## 01 MOTIVATION AND OBJECTIVE

The Forgotten Village is a mystery role-playing game in which an amnesiac traveler investigates five village houses, solves engineering puzzles, and collects records of a vanished researcher. After unlocking the sixth, abandoned house, the player reconstructs an unsafe experiment and discovers that the missing researcher is the player.

The project extends this narrative into an interactive VR–physical system. A real control console becomes part of the fictional experiment. Our goal is to connect the virtual investigation to the physical world: real-world objects become verifiable clues, and a tangible archive chest delivers the final identity reveal.

## 02 EXEPECTED RESULTS

We plan to deliver a six-house VR mystery, five engineering puzzles, five camera-verified real-world object quests, a clue journal, and a completed finale. The chest will open only after all clue flags and the final investigation are complete.

## 03 GAME CONCEPT AND ENGINEERING CONTENT

The first five houses may be explored in any order. In each house, the player first solves an engineering puzzle; an NPC then requests a real-world object. The player removes the Quest headset, finds the object, and presents it to a fixed Raspberry Pi camera. A YOLO-based detector checks the object category, captures evidence, and sends a result to Unity. The player returns to VR to receive the corresponding clue.

- Elder — voltage-divider loading; find a book (research notes).
- Alchemist — RC low-pass filter; find a bottle (sample container).
- Carpenter — MOSFET amplifier gain; find a computer mouse (workstation tool).
- Technician — C++ object slicing; find a cell phone (diagnostic record).
- Clockmaker — VLSI setup timing slack; find a clock (experiment timeline).

After all five puzzle-and-object tasks are completed, the abandoned house unlocks. The player compares archived source code, deployed-build records, and runtime logs to establish the unsafe software decision while treating timing slack as a separate potential risk.

## 04 HARDWARE AND VR SYSTEM CONFIGURATION

Meta Quest runs the Unity PC VR experience through Quest Link. A Windows PC exchanges recognition results with Raspberry Pi over the local network. A Raspberry Pi camera serves as the physical sensor; a lightweight YOLO detector identifies one of the five designated object categories and saves a successful snapshot. Unity communicates with Arduino Uno over USB Serial. When the final VR investigation is completed, Unity sends an unlock command; Arduino drives an SG90 servo to release the latch of a small physical archive chest (actuator).

Player interaction: VR puzzle → remove headset → camera-based object verification → return to VR → clue collected. Final actuation: final puzzle completed → Unity command → Arduino → chest latch released. A manual release and supervised reset will be provided.




## 05 INNOVATION AND PROMOTION POINTS

- Reality as a clue source: players must identify familiar objects beyond the virtual game environment.
- Logical integration: Raspberry Pi sensing updates Unity quest state; Unity then commands a physical Arduino actuator.
- Engineering mystery: circuits, signal processing, analog electronics, advanced OOP, and VLSI form one evidence-based narrative.
- Tangible ending: the servo-operated chest reveals a student ID and letter, confirming that the player is the missing researcher..

## 06 TEAM MEMBERS AND RESPONSIBILITIES

**程婕茵 — Player Development & Gameplay Design**

- Develop player movement, VR interactions, and object manipulation.
- Design and implement three houses
- Integrate real-world object recognition results into VR gameplay and clue collection.

**范芷祺 — UI Systems & Hardware Integration**

- Develop the dialogue system, inventory, clue journal, and quest UI.
- Design and implement the remaining three houses.
- Implement Raspberry Pi camera recognition, YOLO inference, Unity communication, and Arduino-controlled servo chest.

## 07 DISCLOSURE OF AI ASSISTANCE

ChatGPT was used to assist with proposal writing, and language refinement. AI-generated images were used to illustrate the expected gameplay. The team will independently implement, integrate, and test the proposed hardware and software system.

## 08 REFERENCE

Unity Documentation — https://docs.unity3d.com/

Meta Quest Developer Documentation — https://developers.meta.com/horizon/

Arduino Documentation — https://docs.arduino.cc/